Fiscal Year:	FY 2008	**Task Last Updated:**	FY 09/03/2008
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Project Title:	Non-Adrenergic Mechanisms of Cardiovascular Deconditioning		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Cardiovascular Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	03/01/2004	**End Date:**	02/29/2008
No. of Post Docs:	8	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	8	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Berkowitz, Dan (The Johns Hopkins University) Hare, Joshua (The Johns Hopkins University) Nyhan, Daniel (The Johns Hopkins University)		
Grant/Contract No.:	NCC 9-58-CA00405		
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Performance Goal Text:			

Task Description:	While our previous focus had remained on altered adrenergic signaling (from baroreceptor to adrenergic receptor) in the patho-physiology of orthostatic intolerance, the potential role of nitric oxide, a critical modulator of both cardiac and vascular function became the focus of our investigation. In the vasculature, we have identified novel mechanisms by which NO regulates vascular endothelial function. We have also demonstrated that enhanced NO dependent signaling (as a result of NOS phosphorylation) contributes to vascular contractile hyporesponsiveness, a central component of orthostasis. We are currently using in-vivo techniques to non-invasively and serially measure vascular stiffness (as measured by pulse wave velocity) and determine the contribution of altered arterial compliance to OI. In addition, one of our graduate students, Mr. Eric Tuday, was selected as a summer intern at the Johnson Space Center, in Drs. Janice Meck's and Steven Platt's Cardiovascular Laboratories. His collaborative studies with our and Dr. Platt's group has proven to be very fruitfull and we will be determining the changes in aortic and arterial compliances before, during and after long term bed rest in human volunteers. These studies test the hypothesis that long term bed rest increases arterial compliance which subsequently can decrease the cardiac output response during orthostatic challenges. With regard to the heart, we have determined for the first time that it is not loss of cardiac mass, but altered contractile function that contribute to OI. This depressed contractile function is likely the result of altered adrenergic signaling (of which we have identified a novel arm of the beta-l beta-2 AR signaling pathway) or altered NO signaling. We have made significant contributions to the understanding of the intricate molecular mechanisms underlying the role of NO in the modulating cardiac contractile structure and function in general. We continue to determine the role of dysregulated NO in the pathobiology of orthostatic intolerance. We now have a greater appreciation for the NO signaling dysregulation in 0I and are uniquely poised to test the efficacy of the countermeasures aimed at regulating NO as laid out in our original proposal. We currently have a patent on the target site to inhibit Arginase II. We are currently working with a venture capital group to further fund work to develop a specific designer drug which will inhibit Arginase II. The result of these negotiations has been very positive indeed. Red Abbey, a venture capital fund, is securing licensing through Acidophil, LLC. Acidophil, LLC is an intellectual property development company whose goal is to translate innovative life science technologies into products and businesses. Acidophil’s founders are Nobel Prize winner Sydney Brenner MD, Ph.D., and Philip Goelet Ph.D. Licensing will be from both The Johns Hopkins University and the University of Pennsylvania, both of which are necessary to take this technology into the market place. The new company's name will be Arginetix, Inc., located at 2330 West Joppa Road, Suite 330, Lutherville, MD 21093. Truly this entire project has been translational and has the potential to advance novel targeted drugs to the clinical environment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	These findings could have important therapeutic significance for both astronauts and long-term bed rest patients. Both of these groups are predisposed to orthostatic intolerance upon returning to normal gravity and returning to the upright position, respectively. These findings could also have important therapeutic significance for the normal aging population. In our animal models, we have used aortic atherosclerosis and plaque formation as the target endpoint, as this is an accessible site for the study of the vascular changes. Since it is will accepted that dysfunctional endothelium is the root cause of initiating the atherosclerotic process, ArgII would promise to be a target for all sites of atherosclerosis. In hypercholesteremia, supplementation of L-arginine improves endothelial-dependent relaxation and NO production. Arg activity has also been shown to be increased in diabetes and hypertension induced by various cytokines and inflammatory stimuli. There is an emerging body of data that ArgII is up-regulated in the penis in humans with diabetes. Inhibition of Arg has a significant enhancing effect on erectile function in aging and other disease processes in which NO signaling may be dysregulated, such as diabetes. Although it has not been tested as yet it is likely that diabetes may be associated with an increase in endothelial Arg activity, furthermore, Arg may well be a target for the treatment of diabetes-associated endothelial dysfunction. This can be easily tested in a diabetic mouse or rat model in the future. We currently have four patents pending on the target sites to inhibit Arginase II. Our group of investigators has been working with a venture capital group to further fund the work described in this Project Report to develop a specific drug which will inhibit Arginase II. The result of these negotiations has been very positive indeed. Red Abbey Venture Partnerships, a life sciences private equity partnership, is securing licensing through Acidophil, LLC. Acidophil, LLC is an intellectual property development company whose goal is to translate innovative life science technologies into products and businesses. Acidophil’s founders are Nobel Prize winner Sydney Brenner MD, Ph.D., and Philip Goelet Ph.D. Licensing will be from both The Johns Hopkins University and the University of Pennsylvania, both of which are necessary to take this technology into the market place. The new company's name will be Arginetix, Inc., located at 2330 West Joppa Road, Suite 330, Lutherville, MD 21093. Truly this enite project has been translational and has the potential to advance novel targeted drugs to the clinical environment and a broad spectrum of patients.
Task Progress:	1. The role of the endothelium and the NO/cGMP pathway in vascular hypo-responsiveness and remodeling in microgravity is nearly completed and will be finalized with manuscripts mid-year of Year 3 of this proposal. 2. The role of NO in modulating myocardial contractility and hypertrophy has just begun and we have finished the first of a series of studies during Year 3, this coming year. 3. Whether up-regulation of the NO/cGMP system results in decreased vascular and ventricular stiffness as measured by integrated cardiovascular function and ventricular-vascular coupling will begin toward the end of Year 3 of this project. We have delayed working on this aspect of the project since it relies on using our chronic rat preparation. This preparation requires that we purchase a microvascular telemetry system to monitor the rats on a 24-hour basis. 4. We have been able to show the dysregulation in vascular smooth muscle currently has a target site by which this dysregulation occurs. Our data demonstrating that endothelial cells isolated from ArgII-/- mice are resistant to ROS production and do not exhibit a decrease in NO production following OxLDL exposure suggests that it is the endothelium that is the source of Arg and thus, the target for Arg inhibition. 5. In animal models, we have used aortic atherosclerosis and plaque formation as the target endpoint, as this is an accessible site for the study of the vascular changes. Since it is well accepted that dysfunctional endothelium is the root cause of initiating the atherosclerotic process, ArgII would promise to be a target for all sites of atherosclerosis. 6. We have sought additional funding to develop a specific inhibitor of ArginaseII. This search has resulted in a partnership between The Johns Hopkins University, the University of Pennsylvania and Acidophil, LLC to start a new company. This company will develop new specific blockers of Arginase II and bring these products to the market place.

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